

Financial Econometrics

Experiments, Quasi-experiments, and DiD

Maximilian Rohrer

Overview

- ▶ Basics of experiments and quasi-experiments
- ▶ Problems with non-random treatment
- ▶ Experiment
- ▶ Quasi-experiment
- ▶ Difference-in-Difference estimator
- ▶ Construction of control group

Basics of experiments and quasi-experiments

Experiments: Why?

- ▶ Ideal randomized controlled experiments provide a benchmark for assessing observational studies
- ▶ Actual experiments are rare (\$\$\$) but influential
- ▶ Experiments can solve the threats to internal validity of observational studies, but they have their own threats to internal validity
- ▶ Thinking about experiments helps us to understand quasi-experiments, or “natural experiments”

Terminology

- ▶ An **experiment** is designed and implemented consciously by human researchers. An experiment entails conscious use of a treatment and control group with random assignment (e.g. clinical trials of a drug).
- ▶ A **quasi-experiment** or **natural experiment** has a source of randomization that is “as if” randomly assigned, but this variation was not part of a conscious randomized treatment and control design.

Problems with non-random treatment

Do hospitals make people healthy?

Group	Sample Size	Mean health status	Std. Error
Hospital	7774	2.79	0.014
No Hospital	90049	2.07	0.003

Figure 1: National Health survey 2005; Health 1 (good) to 5 (poor)

More formally...

- ▶ $D_i = \{0, 1\}$ = hospital treatment for individual i
- ▶ Y_i = health outcome
- ▶ For any individual, there are two potential outcomes:

$$Y_i = \begin{cases} Y_{1i} & \text{if } D_i = 1 \\ Y_{0i} & \text{if } D_i = 0 \end{cases}$$
$$= Y_{0i} + (Y_{1i} - Y_{0i}) * D_i$$

- ▶ We are interested in the causal effect of hospitalization for an individual ($Y_{1i} - Y_{0i}$)

More formally...

- ▶ let us do a naive comparison of averages by hospitalization

$$\underbrace{E(Y_{1i}|D_i = 1) - E(Y_{0i}|D_i = 0)}_{\text{Observed difference in average health}} =$$

$$\begin{aligned} & \underbrace{E(Y_{1i}|D_i = 1) - E(Y_{0i}|D_i = 1)}_{\text{Average treatment effect on the treated}} \\ & + \underbrace{E(Y_{0i}|D_i = 1) - E(Y_{0i}|D_i = 0)}_{\text{Selection bias}} \end{aligned}$$

Random assignment solves the selection problem

- ▶ Random assignment implies that D_i is independent of the outcome
- ▶ This implies $E(Y_{0i}|D_i = 0) = E(Y_{0i}|D_i = 1)$ which limits the previous equation to

$$\underbrace{E(Y_{1i}|D_i = 1) - E(Y_{0i}|D_i = 0)}_{\text{Observed difference in average health}} =$$

$$\underbrace{E(Y_{1i}|D_i = 1) - E(Y_{0i}|D_i = 1)}_{\text{Average treatment effect on the treated}}$$

Regression to estimate treatment effect

- ▶ Assuming that treatment is constant across individuals $\rho = Y_{1i} - Y_{0i}$, we can rewrite the treatment effect

$$Y_i = Y_{0i} + (Y_{1i} - Y_{0i}) * D_i$$

as

$$Y_i = \alpha + \rho D_i + u_i$$

- ▶ α will estimate the average outcome without treatment $E(Y_{0i})$
- ▶ Per definition, the error term will include the potential selection bias $u_i = Y_{0i} - E(Y_{0i})$.
- ▶ If there is a selection bias $E(Y_{0i}|D_i = 0) \neq E(Y_{0i}|D_i = 1)$, $E(u_i|D_i) \neq 0$ and zero conditional mean assumption is not satisfied

Experiments

Idealized Experiments

- ▶ An ideal randomized controlled experiment randomly assigns subjects to treatment and control groups
- ▶ More generally, the treatment level x is randomly assigned:
$$y_i = \beta_0 + \beta_1 x_i + u_i$$
- ▶ If x is randomly assigned (for example by computer) then u and x are independently distributed and $E(u|x) = 0$, so OLS yields an unbiased estimator of β_1
- ▶ The causal effect is the population value of β_1 in an ideal randomized controlled experiment
- ▶ If x is binary, then the causal effect is the difference in means between treatment ($x = 1$) and control group ($x = 0$)

Example: Class size

- ▶ Project STAR (Student-Teacher Achievement Ratio)
- ▶ 4-year study, \$12 million
- ▶ Upon entering the school system, a student was randomly assigned to one of three groups:
 - ▶ regular class (22 - 25 students)
 - ▶ regular class + aide
 - ▶ small class (13 - 17 students)
- ▶ Regular class students re-randomized after first year to regular or regular + aide
- ▶ y = Stanford Achievement Test scores

Threats to Internal Validity ($\text{cov}(x, u) \neq 0$)

- ▶ Failure to randomize (or imperfect randomization)
 - ▶ E.g., allocation of kids to different class sizes on first-come, first-serve basis
- ▶ Failure to follow treatment protocol (or “partial compliance”)
 - ▶ some controls get the treatment
 - ▶ some “treated” get controls
- ▶ Attrition
 - ▶ some subjects drop out
- ▶ Experimental effects:
 - ▶ subject behavior might be affected by being in an experiment (Hawthorne effect)
- ▶ Small sample bias:
 - ▶ Experiments are expensive which often results in a small sample

Example: Class size

- ▶ Partial compliance:
 - ▶ 10% of students switched treatment groups because of “incompatibility” and “behavior problems” - how much of this was because of parental pressure?
 - ▶ New comers: incomplete receipt of treatment for those who move into district after grade 1
- ▶ Attrition:
 - ▶ students move out of district
 - ▶ students leave for private/religious schools

Threats to External Validity

- ▶ Non-representative sample
- ▶ Non-representative program or policy
- ▶ General equilibrium effects (effect of a program can depend on its scale and duration)

Example: Class size

- ▶ The “differences-in-means” regression:

$$y = \beta_0 + \beta_1 \textit{SmallClass} + \beta_2 \textit{RegAide} + \sum_j \beta_j w_j + u$$

- ▶ $\textit{SmallClass} = 1$ if in a small class.
- ▶ $\textit{RegAide} = 1$ if in regular class with aide.
- ▶ Additional regressors (w): teacher experience, free lunch eligibility, gender, race, etc.

Packages you will need today

```
# relevant packages  
require(stargazer) # regression table  
require(lmtest) # robust SE  
require(sandwich) # robust SE
```

Example: Class size

```
# load data
load("STAR.Rdata")

# run regressions
fit1 <- lm(tscorek ~ sck + rak, data=STAR)
robust_se1 <- coeftest(fit1,vcov=vcovHC(fit1))[,2]

fit2 <- lm(tscore1 ~ sc1 + ra1, data = STAR)
robust_se2 <- coeftest(fit2, vcov = vcovHC(fit2))[,2]

fit3 <- lm(tscore2 ~ sc2 + ra2, data = STAR)
robust_se3 <- coeftest(fit3,vcov = vcovHC(fit3))[,2]

fit4 <- lm(tscore3 ~ sc3 + ra3, data = STAR)
robust_se4 <- coeftest(fit4,vcov = vcovHC(fit4))[,2]
```

Example: Class size

```
# display the results
stargazer(fit1, fit2, fit3, fit4,
          se = list(robust_se1, robust_se2,
                    robust_se3, robust_se4),
          type="text",
          keep.stat=c("n", "rsq", "adj.rsq"),
          no.space = T,
          report=('vc*t'))
```

Example: Class size

[illegible]

Example: Class size

```
# Compute standard deviation for all dependent variables  
apply(STAR[,3:6], 2, sd, na.rm = T)
```

```
## tscorek tscore1 tscore2 tscore3  
## 73.74660 91.28273 84.07682 73.26685
```

```
# rescale all dependent variables (Mean 0, StdDev 1)
```

```
STAR$tscorek.sc <- scale(STAR$tscorek)  
STAR$tscore1.sc <- scale(STAR$tscore1)  
STAR$tscore2.sc <- scale(STAR$tscore2)  
STAR$tscore3.sc <- scale(STAR$tscore3)
```

```
# Compute standard deviation for all dependent variables  
apply(STAR[,19:22], 2, sd, na.rm = T)
```

```
## tscorek.sc tscore1.sc tscore2.sc tscore3.sc  
##          1          1          1          1
```

Example: Class size

```
# run regressions
fit1 <- lm(tscorek.sc ~ sck + rak, data=STAR)
robust_se1 <- coeftest(fit1,vcov=vcovHC(fit1))[,2]

fit2 <- lm(tscore1.sc ~ sc1 + ra1, data = STAR)
robust_se2 <- coeftest(fit2, vcov = vcovHC(fit2))[,2]

fit3 <- lm(tscore2.sc ~ sc2 + ra2, data = STAR)
robust_se3 <- coeftest(fit3,vcov = vcovHC(fit3))[,2]

fit4 <- lm(tscore3.sc ~ sc3 + ra3, data = STAR)
robust_se4 <- coeftest(fit4,vcov = vcovHC(fit4))[,2]
```


Example: Class size

```
##
## =====
##                               Dependent variable:
##                               -----
##          tscorek.sc tscore1.sc tscore2.sc tscore3.sc
##                (1)      (2)      (3)      (4)
## -----
## sck          0.188***
##              t = 5.662
## rak          0.004
##              t = 0.138
## sc1          0.326***
##              t = 10.516
## ra1          0.131***
##              t = 4.508
## sc2          0.231***
##              t = 7.150
## ra2          0.041
##              t = 1.367
## sc3          0.213***
##              t = 6.503
## ra3          -0.004
##              t = -0.128
## Constant    -0.058*** -0.137*** -0.083*** -0.066***
##              t = -2.623 t = -6.992 t = -3.856 t = -2.894
## -----
## Observations  5,786      6,379      6,049      5,967
## R2            0.007      0.017      0.009      0.010
## Adjusted R2   0.007      0.017      0.009      0.010
## =====
## Note:                *p<0.1; **p<0.05; ***p<0.01
```

Example: Class size

```
# run regressions: Adding controls
fit1 <- lm(tscorek.sc ~ sck + rak,data=STAR)
robust_se1 <- coeftest(fit1, vcov=vcovHC(fit1))[,2]

fit2 <- lm(tscorek.sc ~ sck + rak + totexpk,
           data = STAR)
robust_se2 <- coeftest(fit2, vcov=vcovHC(fit2))[,2]

fit3 <- lm(tscorek.sc ~ sck + rak + totexpk + factor(schidkn),
           data = STAR)
robust_se3 <- coeftest(fit3, vcov=vcovHC(fit3))[,2]

fit4 <- lm(tscorek.sc ~ sck + rak + totexpk + boy + freelunk +
           black + other + factor(schidkn),data = STAR)
robust_se4 <- coeftest(fit4, vcov=vcovHC(fit4))[,2]
```

Example: Class size

```
##
## =====
##                               Dependent variable:
##                               -----
##                               tscorek.sc
##                               (1)      (2)      (3)      (4)
## -----
## sck          0.188***   0.190***   0.216***   0.215***
##              t = 5.662   t = 5.722   t = 7.060   t = 7.319
##
## rak          0.004      -0.008      0.016      0.024
##              t = 0.138   t = -0.266   t = 0.593   t = 0.904
##
## totexpk          0.020***   0.010***   0.009***
##              t = 8.672   t = 4.346   t = 3.964
##
## boy                      -0.164***
##                          t = -7.175
##
## freelunk          -0.471***
##                  t = -17.331
##
## black             -0.345***
##                  t = -7.205
##
## other             -0.115
##                  t = -0.649
##
## Constant        -0.058***   -0.239***    0.045    0.323***
##                  t = -2.623   t = -7.919   t = 0.434   t = 3.292
##
## -----
## Observations    5,786      5,766      5,766      5,748
## =====
## Note:                *p<0.1; **p<0.05; ***p<0.01
```

Example: Class size

- ▶ Threads to internal validity?
 - ▶ E.g., partial compliance/incomplete treatment
 - ▶ Solution: use 2SLS with an instrument $z = \text{initial assignment}$
 - ▶ Krueger (1999) shows that 2SLS and OLS estimates are similar, so this bias seems not to be large
- ▶ Main findings:
 - ▶ The effects of being in a small class are substantial
 - ▶ The effects are comparable to having a 20-year veteran as a teacher rather than a new teacher
 - ▶ However the effect does not cumulate or increase while moving to an older grade

Quasi-experiments

Quasi-experiment

- ▶ A quasi experiment or natural experiment has a source of randomization that is “as if” randomly assigned, but the variation was not part of a conscious randomized treatment and control design
- ▶ Two cases:
 - ▶ Treatment (d) is “as if” randomly assigned
 - ▶ A variable (z) that influences treatment (d) is “as if” randomly assigned: use IV regressions

Example of quasi-experiments

- ▶ NYSE Autoquote
 - ▶ Technological change in market design: best bid-offer is updated automatically rather than manually
 - ▶ Implementation in a staggered way
 - ▶ Increase speed and automation
- ▶ Algo trading
 - ▶ Algos react beyond human capabilities: e.g. in 2020 NASDAQ upgraded to nanosecond level

Quasi-experiment

- ▶ If the treatment in a quasi-experiment is “as if” randomly assigned conditional on observed characteristics
- ▶ ... we can estimate the unbiased treatment effect by OLS while including w as control variables: $Y_i = \alpha + \rho D_i + \beta w_i + u_i$
- ▶ What if the treatment in a quasi-experiment is “as-if” randomly assigned conditional on unobserved characteristics?
- ▶ If these differences in unobserved characteristics are time-invariant, then use difference-in-difference approach
 - ▶ treatment and control group
 - ▶ before and after treatment

Quick Recap

- ▶ Basics of experiments and quasi-experiments
- ▶ Problems with non-random treatment
- ▶ Experiment
- ▶ Quasi-experiment
- ▶ Difference-in-Difference estimator
- ▶ Construction of control group

Difference-in-Difference estimator

Difference in difference

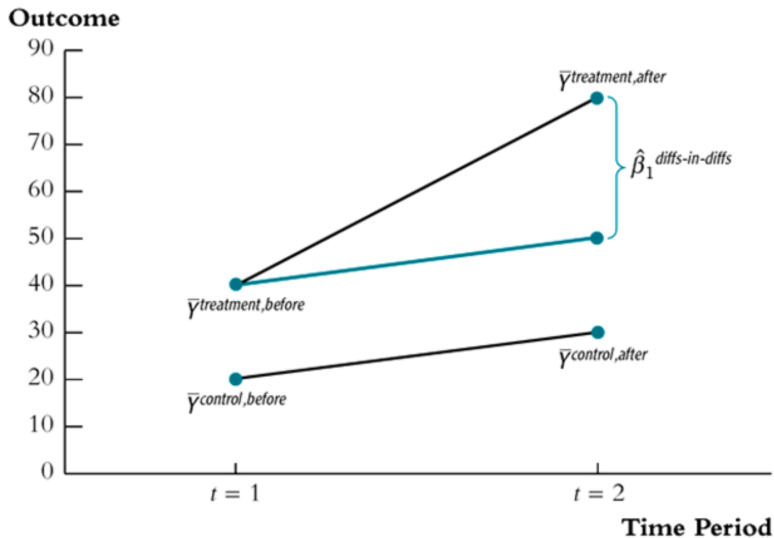


Figure 2: Difference in Difference

Difference in difference

$$\Delta Y_i = \alpha + \beta_1 G_i + u_i$$

- ▶ $\Delta Y_i = Y_i^{after} - Y_i^{before}$
- ▶ Treatment group G_i : turns one if for treatment group and zero for control group
- ▶ β_1 is the difference in difference (DiD) estimator

Difference in difference: panel data formulation

$$y_{i,t} = \alpha + \beta_1 D_t * G_i + \beta_2 D_t + \beta_3 G_i + u_{i,t}$$

- ▶ Treatment dummy D_t : turns one after treatment and zero before
- ▶ Treatment group G_i : turns one if for treatment group and zero for control group
- ▶ Interaction term $D_t * G_i$: effect of being in the treatment group after treatment was received
- ▶ β_1 is the difference in difference (DiD) estimator

Parallel trend assumption

- ▶ Parallel trend assumption requires that in the absence of treatment, the difference between the treatment and control group is constant over time.
- ▶ We cannot test that, but
 - ▶ If the treatment and control firms seem similar before the treatment, this is more likely to be the case.

Construction of control group

Hypothetical example

“BEST IS”, an ice cream store in Bergen, starts using a new expensive machine that makes even better ice cream. They start using this great machine on 20th of June 2019. The weather on that day is really nice but in contrast to the previous day no cruise ship arrives in Bergen.

You are interested in how much the increase in quality of ice cream increased the number of ice creams sold. You recognise that you can use quasi-experimental techniques to study this question, in particular DiD.

However, what ice cream store(s) is/are the best control group for “BEST IS”?

Hypothetical example



Considerations

- ▶ Parallel trend
 - ▶ Weather (A) to (C)
 - ▶ Cruise ship (A) to (B)
- ▶ Does the treatment affect control firms indirectly?
 - ▶ Decreased demand for (A) and (B)
 - ▶ Lower prices or investment in good machine for (A) to (C)

Solutions: matching

Good control group should satisfy “parallel trends” assumption

For each treated firm, we select one or more control firms. Two methodologies:

1. Nearest neighbour
2. Propensity score

Nearest neighbour

- ▶ For each treatment unit, we select a control firm that has the smallest matching error
- ▶ Matching error: let's assume we want to match based on market capitalization *mcap* and price *prc*

$$\text{Matching error}_{i,j} = \frac{|mcap_i/mcap_j - 1| + |prc_i/prc_j - 1|}{2}$$

Propensity score

- ▶ We model the probability of being treated with a logit model
 - ▶ Dependent variable: dummy that turns one if the firm is treated
 - ▶ Independent variable: matching characteristics
- ▶ For example: we match based on market capitalization (MCAP) and price (PRC)

$$Prob(y = 1) = \Lambda(\beta_0 + \beta_1 MCAP + \beta_2 PRC)$$

- ▶ Subsequently, we select for each treated firm a control firm with a similar predicted probability of being treated

Matching procedure

- ▶ Different purpose and matching function
 - ▶ Nearest neighbour
 - ▶ Propensity score
- ▶ Selection from pool of potential control firms
 - ▶ Replacement or not
 - ▶ Minimal distance per pair versus minimal aggregate distance
 - ▶ Number of matched firms
- ▶ Do observable or unobservable characteristics define treatment?

Summary

- ▶ Basics of experiments and quasi-experiments
- ▶ Problems with non-random treatment
- ▶ Experiment
- ▶ Quasi-experiment
- ▶ Difference-in-Difference estimator
- ▶ Construction of control group

Textbook

- ▶ Experiments and Quasi-experiments: Chapter 13, Stock and Watson, Introduction to Econometrics, Global Edition, 4th edition